RUBEN DARIO FLOREZ ZELA

Electronic Engineer · Human-Centered AI for Intelligent Vehicles · Computer Vision & Embedded AI

rubendfz2206@gmail.com | Cusco, Peru

ORCID | Google Scholar | LinkedIn | GitHub | Personal Web

RENACYT Level V (CONCYTEC) · Google Scholar: 221 citations · h-index 5 · i10-index 3 (Sep. 2026)

RESEARCH PROFILE

Electronic engineer and AI researcher working at the intersection of human-centered artificial intelligence, computer vision, embedded systems, and automated driving. My research spans driver monitoring, robust and explainable perception, conformal risk control for SAE Level 2 automated driving, and deployment-aware evaluation of vision models on embedded hardware. I am particularly interested in reliable perception and decision-support methods for ADAS and intelligent vehicles under real-world uncertainty and domain shift.

EDUCATION

M.Sc. in Electronic Engineering (Automation & Instrumentation) — Studies completed Jul. 2026
Universidad Nacional de San Agustín de Arequipa (UNSA), Arequipa, Peru

- **48 credits completed; weighted average: 17.10/20** (official passing grades: 13–20). Degree expected 2026.
- **M.Sc. thesis (in progress):** *Multi-Horizon Prediction of Delayed-Response Risk and Three-State Conformal Supervision for Longitudinal Fallback in SAE Level 2 Automated Driving.*

Bachelor's Degree in Electronic Engineering — Degree awarded Feb. 2023
Universidad Nacional de San Antonio Abad del Cusco (UNSAAC), Cusco, Peru

- 223 approved academic credits; relevant coursework includes Artificial Intelligence, Robotics, Digital Signal Processing, Control Theory, Industrial Control Systems, Instrumentation, and Microprocessors and Computer Architecture.

Professional Title: Electronic Engineer — Awarded Feb. 2024
UNSAAC, Cusco, Peru | Professional License (CIP): 336757 | Obtained by thesis

- **Professional thesis:** *Design and Implementation of a Real-Time Driver Drowsiness Detection System Using Computer Vision and Convolutional Neural Networks*; **unanimously approved with excellence (19/20)**.

SELECTED RESEARCH EXPERIENCE

Safety-Critical Risk Control for Automated Driving (M.Sc. Research) — 2025 – 2026
Universidad Nacional de San Agustín de Arequipa (UNSA)

- Developed driver-grouped machine-learning models for multi-horizon prediction of delayed-response risk in SAE Level 2 automated driving.
- Designed calibrated and conformal risk-control mechanisms for graded longitudinal intervention with HUMAN–ASSIST–FALLBACK supervision.
- Evaluated safety effects through counterfactual longitudinal simulation under driver-specific uncertainty.

Explainable AI Auditing for Safety-Critical Perception (Lead Researcher) — 2026
Independent research

- Developed a confidence-controlled framework for auditing pedestrian-detector explanations under cross-domain distribution shift.
- Evaluated explanation faithfulness using detector-specific perturbation methods while controlling detector confidence as a potential confounding factor.
- Resulting work accepted as a Regular Paper at **IEEE ICVES 2026**.

Deployment-Aware Vision Benchmarking for Driver Monitoring (Lead Researcher) — 2026 – Present
Independent research / public reproducibility repository

- Benchmarking lightweight CNN and transformer-based vision models under clean, corrupted, and cross-domain conditions.
- Evaluating embedded deployment using NVIDIA Jetson Nano and TensorRT with latency, throughput, memory, and reliability criteria.
- Maintaining reproducible protocols, frozen model semantics, experiment manifests, and audit-oriented evaluation pipelines.

Embedded Vision for Driver-State Monitoring (Lead Researcher) — 2023
LIECAR Lab, UNSAAC

- Developed CNN-based real-time driver monitoring using eye and facial-region analysis and deployed real-time inference on NVIDIA Jetson Nano.
- Designed ROI correction procedures robust to illumination variability.

SELECTED RESEARCH OUTPUTS

Accepted peer-reviewed conference paper

[1] **Florez-Zela, R. D.** *Confidence-Controlled XAI Auditing for Pedestrian Detection under Domain Shift.* Accepted as a Regular Paper for the *2026 IEEE International Conference on Vehicular Electronics and Safety (ICVES).*

Current journal manuscript and preprint

[1] **Florez, R. D.** *Driver-Grouped Conformal Risk Control for Graded Longitudinal Intervention in Level 2 Automated Driving: A Counterfactual Simulation Study.* Manuscript submitted to *IEEE Transactions on Intelligent Vehicles*, 2026.

[2] **Florez, R. D.** *Beyond Aggregate Scores: Deployment-Aware and Non-Compensatory Benchmarking of Vision-Based Eye-State Recognition Models for Driver Monitoring.* arXiv preprint 2606.08123, 2026.

Peer-reviewed journal articles

[1] **Florez, R.**, Palomino-Quispe, F., Alvarez, A. B., Coaquira-Castillo, R. J., & Herrera-Levano, J. C. (2024). A Real-Time Embedded System for Driver Drowsiness Detection Based on Visual Analysis of the Eyes and Mouth Using Convolutional Neural Network and Mouth Aspect Ratio. *Sensors*, 24(19), 6261.

[2] **Florez, R.**, Palomino-Quispe, F., Coaquira-Castillo, R. J., Herrera-Levano, J. C., Paixão, T., & Alvarez, A. B. (2023). A CNN-Based Approach for Driver Drowsiness Detection by Real-Time Eye State Identification. *Applied Sciences*, 13(13), 7849.

Selected peer-reviewed conference paper

[1] Paixão, T., Alvarez, A. B., **Florez, R.**, & Palomino-Quispe, F. (2023). Fuzzy Control for Simplified Lumbar Spine Robotic Mechanism Motion. *2023 IEEE International Conference on Networking, Sensing and Control (ICNSC)*, pp. 1–6. IEEE.

Full publication list: Google Scholar

ACADEMIC POSITIONS

Contract Lecturer – Electronic Engineering Department Aug. 2024 – Jan. 2026
Universidad Nacional de San Antonio Abad del Cusco (UNSAAC), Cusco, Peru

- Taught Artificial Intelligence, Digital Image Processing, Robotics, and Electronics Laboratories; supervised applied student work in machine learning, computer vision, and embedded systems.

Teaching Assistant – Electronic Engineering Department Apr. 2024 – Aug. 2024
Universidad Nacional de San Antonio Abad del Cusco (UNSAAC), Cusco, Peru

- Supported practical and laboratory sessions in electronics and AI-related courses.

TECHNICAL SKILLS

Programming:	Python, C/C++, MATLAB
Machine Learning:	PyTorch, TensorFlow, Keras, scikit-learn; classification, calibration, uncertainty-aware evaluation
Computer Vision & XAI:	OpenCV, RGB/NIR imaging, gaze analysis, facial landmarks, pedestrian detection, explanation auditing
Embedded & Edge AI:	NVIDIA Jetson Nano, TensorRT, CUDA-enabled inference, Raspberry Pi, STM32, ESP32
Control & Real-Time:	Digital control, PID/fuzzy control, real-time automation, instrumentation
Research Software:	Linux, Git/GitHub, reproducible experiment organization, deterministic evaluation workflows

RESEARCH INTERESTS

· Human-Centered AI · Driver Monitoring · Intelligent Vehicles · Automated Driving · Robust & Explainable Perception · Computer Vision · Embedded & Edge AI · Safety-Critical Machine Learning · Intelligent Transportation Systems

SELECTED DISTINCTIONS, SERVICE & LANGUAGES

- **RENACYT Level V Researcher**, CONCYTEC, Peru; IEEE Member (ITSS and VTS).
- **1st Place**: VIII Regional Science, Technology & Innovation Fair (2023) and II International Congress of Research, Innovation & Entrepreneurship, UNSAAC (2022).
- Reviewer for *Scientific Reports*, *Scientific Data*, and *ACM Transactions on Intelligent Systems and Technology.*
- **Languages:** Spanish (native); English (intermediate, working proficiency for academic research and scientific communication).